The evolution of 3-points shooting in NBA and its importance in modern basketball

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Introduction

Since the introduction of the 3-point line in the 1979–80 season, the role of the 3-point shot in the NBA has evolved dramatically. In recent years, teams have increasingly relied on 3-point shooting as a core offensive strategy, with players such as Stephen Curry helping accelerate this trend.

However, the extent and timing of this strategic shift are not always clearly quantified. Understanding **how three-point shooting has evolved over time and how teams and players have adapted their strategies** is essential for evaluating modern offensive trends.

This analysis aims to explore the historical evolution of 3-point shooting in the NBA since its introduction, focusing on changes in **shot volume, efficiency, team strategy, and player behaviour**.

The goal is to identify **key trends and strategic shifts** that have contributed to the modern 3-point focused style of play.

Business Impact

Understanding the evolution of 3-point shooting provides valuable insights for basketball decision-makers, including coaching staff, front offices, and analysts.

Key impacts of this analysis include:

1. **Offensive Strategy Optimization**

By identifying how successful teams have increased their reliance on 3-point shooting, teams can better evaluate whether their offensive strategy aligns with modern trends in the NBA.

2. **Player Recruitment and Development**

Insights into the relationship between **3-point volume and efficiency** can help teams prioritize players who provide the greatest offensive value in today's game.

3. **Competitive Benchmarking**

Teams can compare their shot selection strategies with league trends to determine whether they are **keeping pace with modern offensive strategies**.

4. **Game Strategy and Analytics**

Understanding how the 3-point shot affects scoring efficiency can support **data-driven coaching decisions**, such as lineup construction, shot selection, and offensive schemes.

5. **Long-Term Strategic Planning**

Front offices can use historical trends to evaluate how the game is evolving and anticipate future shifts in offensive play.

General Dataset Information

Files name: Player Totals.csv

Team Totals.csv

Description: Basketball statistics from 1947 to present season.

Datasets details: 33120 Rows & 33 Columns

1908 Rows & 28 Columns

Size: 4,388 KB (4.28 MB)

273 KB

Source: Kaggle – Dataset Link

Data Profile

- Import datasets to SQL Lite.
- Create glossary for stats abbreviations from source web.
- Really clean datasets but examine data for possibles inconsistencies.
- Notable features:
  - 80 seasons of statistics.
  - 97 unique team names.
  - 5384 unique player ids.
  - 5 Players position
  - Player's dataset includes 26 stats columns and Teams dataset includes 23.
- Recorded inconsistencies:
  - NA values in player's position.
  - NA values in percentage stats in players dataset.
  - Players that change teams during season appears multiple times per season, one for each team and another recollecting the sum of stats for the season.
  - Team abbreviations as 2TM, 3TM, 4TM, 5TM for players with more than one team per season.

Data Wrangling

- Deleted the stats from current season (unfinished season) and seasons before 1980 (season where 3-point line were introduced).
  - 8503 rows deleted from player's dataset.
  - 577 rows deleted from team's dataset.
- All the NA values found in the position column belong to season before 1980.
- All the NA values found in the percentage's columns have been changed to 0.
- Removed lg (league) column, only NBA league existed in this period of time.
- To avoid having duplicate players per season, in these cases only the combine row with the season stats is selected, unable to pull this query, Meta AI was consulted.

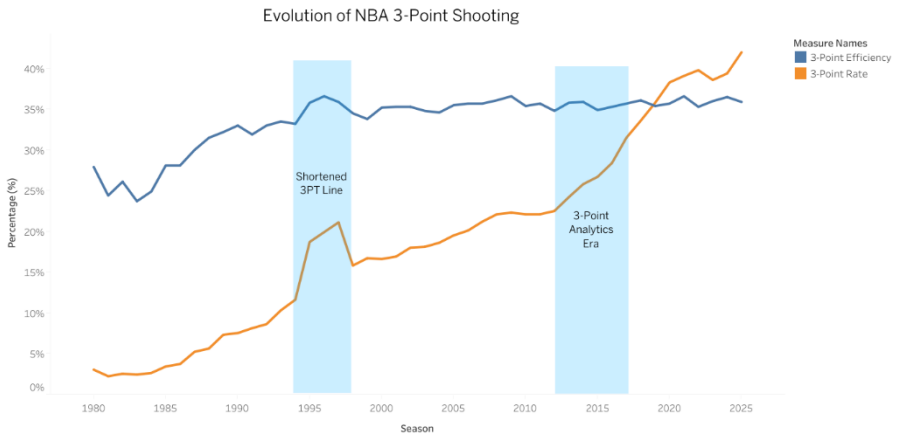
Target Features

The columns that will be used to simplify and analyse the problem statement are:

- season: range of seasons, from 1980 to 2025.
- team: name of teams that have been part of the league during this period of time.
- player: player's names that have been part of the league.
- g: number of games during the season.
- pos: Position in game of the player.
- mp: Minutes played in a season by player.
- x3p: 3-Points made during season by teams and players.
- x3pa: 3-Points attempts during season by teams and players.
- fg: Field Goals made during season by teams and players.
- fga: Field Goals attempts made during season by teams and players.

List of Analysis

1. **League 3-Point Rate and Efficiency over time.**



Since the introduction of the three-point line in the **1979–80 NBA season**, the use of 3-point shots has increased steadily across the league. However, the rate of adoption varied over time, creating distinct eras in the evolution of 3-point shooting.

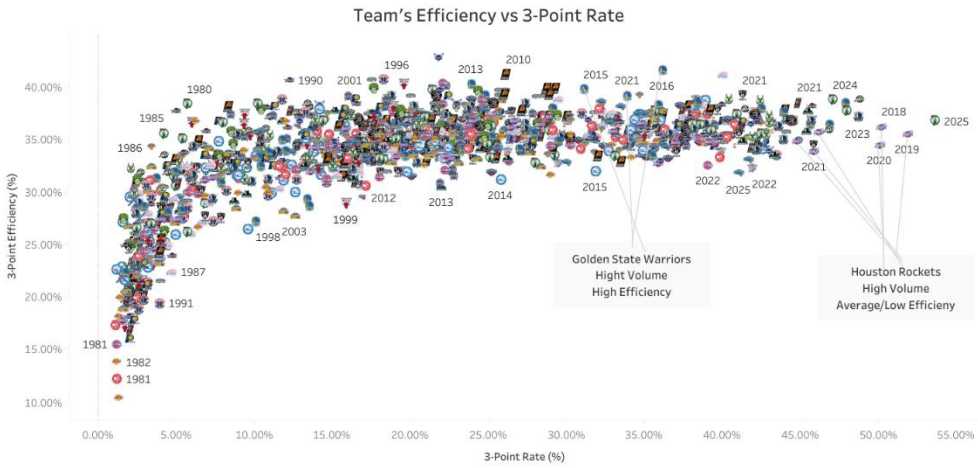
Assimilation Era (1980–early 1990's)
Following its introduction, teams were slow to incorporate the 3-point shot into their offensive strategies. During the first decade, the share of 3-point attempts increased gradually from roughly **3% to about 12% of total Field Goal Attempts**.

Shortened Line Distance (1994–1997)
To encourage perimeter shooting, the NBA temporarily shortened the 3-point distance starting in the **1993–94 season**. This change led to a noticeable spike in 3-point attempts. After the league restored the original distance, usage declined slightly but continued to grow steadily throughout the 2000's.

3-Point Revolution (mid-2010's onward)
The rise of perimeter-focused offenses accelerated in the 2010's, influenced heavily by players like Stephen Curry and Klay Thompson with the Golden State Warriors. Beginning around the **2015–16 season**, the league experienced a sharp increase in 3-point attempts as teams adopted high-volume three-point strategies.

Interestingly, this dramatic rise in shot volume contrasts with **3-point efficiency**, which has remained relatively stable since the early years of adoption, fluctuating mostly between **32% and 37%**.

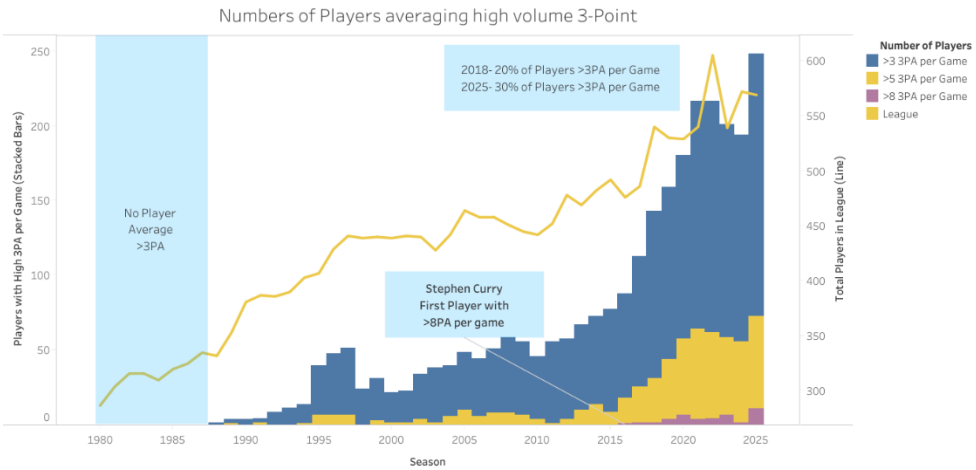
2. **Attempts vs Efficiency by Teams.**



This analysis is consistent with previous observations: in the early years, teams had both **low 3-point usage and low efficiency**. By the mid-1990s, **efficiency stabilized**, with most teams clustered between **32–37%**, while the **volume of 3-point attempts increased** across the league. Notably, some teams emerged as **pioneers of the 3-point strategy**, combining **high volume with high efficiency**, setting the stage for modern perimeter-oriented offenses.

In recent years, the NBA has seen an unprecedented volume of 3-point attempts, with many teams adopting a high-paced, perimeter-focused strategy. Despite the increase in attempts, league-wide 3-point efficiency has plateaued, indicating a balance between volume and accuracy in the modern game.

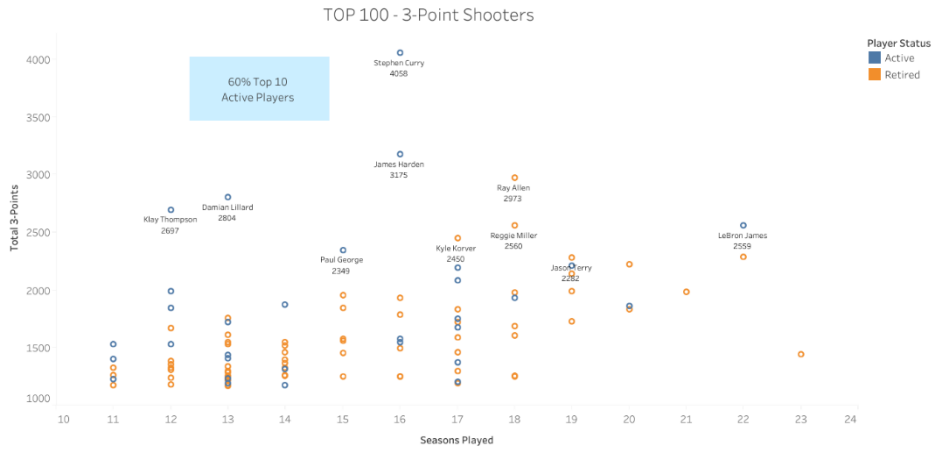
3. **Number of players attempting 3 / 5 / 8 3-Point Shots per game.**



During the **Assimilation Era**, it was rare for any player to average **more than 3 attempts per game** from beyond the 3-point line. The number of players averaging more than 3 attempts increased considerably during the **Shortened Line** era and continued to grow gradually throughout the 2000's, with a few players beginning to **exceed 5 attempts per game**. The **2015–16** season can be considered a turning point: for the first time a player averaged more than **8 three-point attempts per game** (Stephen Curry). Since then, an increasing number of players have joined the group of high-volume 3-point shooters.

This shift reflects the broader transformation of offensive strategies in the modern NBA, where 3-point shooting has become a central component of team offenses.

4. **TOP 100 3-Point Shooters.**



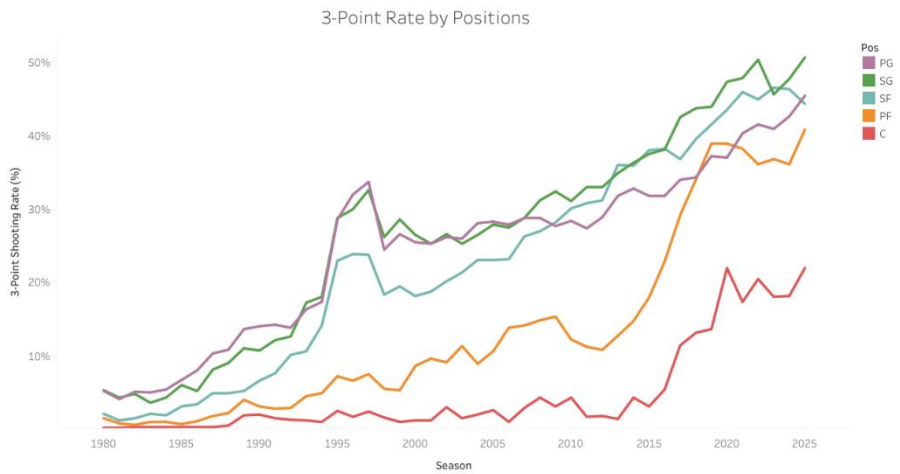
Although only **31% of the Top 100 players in total 3-point field goals are currently active**, these players are disproportionately represented at the top of the leaderboard. Active players account for **21% of the Top 50** and dominate the **Top 10 with 60% of the players still active**.

This pattern reflects the increasing importance of the three-point shot in modern NBA offenses. The concentration of active players among the highest ranks suggests that current players accumulate three-point totals at a much faster pace than players from previous eras.

This trend is largely driven by the modern offensive strategies adopted across the league, where three-point shooting has become a central component of team scoring systems. Players such as Stephen Curry, James Harden, and Damian Lillard exemplify this shift, as they have reached historically high totals while still active.

This distribution suggests that the leaderboard is likely to continue changing rapidly as active players climb the rankings in a league increasingly defined by high-volume three-point shooting.

3-Point Shooting Rate by Position.



Traditionally, 3-point shooting has been associated with the **perimeter positions** in basketball. Point Guards (PG), Shooting Guards (SG), and Small Forwards (SF) have historically taken the majority of shots from beyond the 3-point line. This pattern is reflected in the steady increase in the 3-point rate among

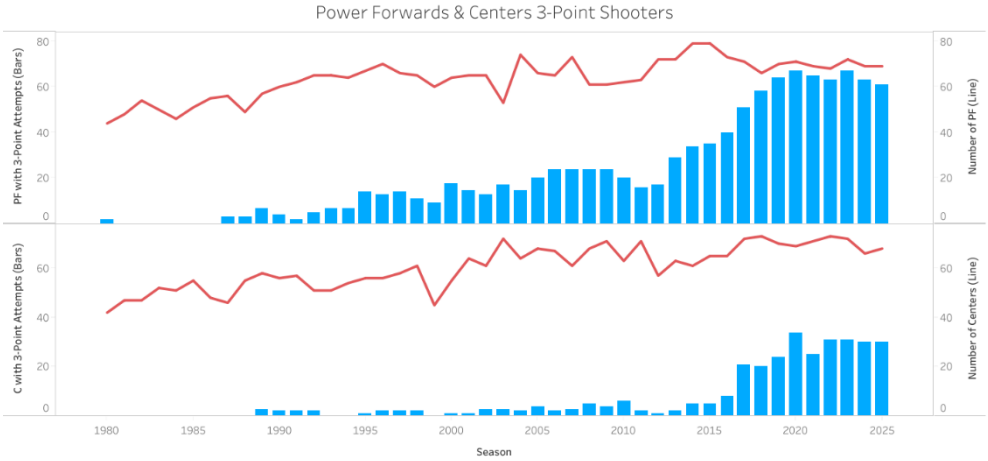
these positions over time, reaching a point where **approximately 50% of the shots taken by Shooting Guards are now 3-point attempts**.

However, the evolution of the 3-point shot is even more evident when examining the **interior positions**. Historically, Power Forwards (PF) and Centers (C) rarely attempted shots from beyond the arc. For many years, the 3-point rate for Power Forwards remained around **10% of their total field goal attempts**, while Centers rarely exceeded **5%**.

In recent years, this pattern has changed dramatically. Modern offensive strategies have increasingly encouraged interior players to stretch the floor and contribute to perimeter shooting. As a result, by the **2024–25 season**, **41% of the shots taken by Power Forwards were 3-point attempts**, while **Centers reached a rate of 22%**.

This shift illustrates how the modern game has redefined positional roles, with even traditionally interior players adapting to a league-wide emphasis on spacing and perimeter shooting.

5. **Power Forwards & Centers with > 1 3PA per game.**



Focusing on the **interior positions**, it is clear that Power Forwards (PF) and Centers (C) were late adopters of the three-point shot. In the early years, **almost no players in these positions attempted even one three-pointer per game**, reflecting the traditional emphasis on interior scoring.

In the 1990s, **Power Forwards began experimenting with perimeter shooting**, with nearly **20% attempting three-pointers more regularly**, while Centers largely remained focused on inside play, with only **1–2% averaging at least one three-pointer per game**.

Over the last decade, however, the trend has shifted dramatically. By the **2024–25 season**, approximately **90% of Power Forwards** and **45% of Centers** are taking at least one three-pointer per game. This represents a major transformation in positional roles, demonstrating how interior players have increasingly adapted to modern, perimeter-oriented offensive strategies.

Conclusion

The analysis confirms that the role of the three-point shot in the NBA has undergone a profound transformation since its introduction in the 1979–80 season. What began as a marginal scoring option has evolved into a central component of modern offensive strategy.

One of the most significant findings is the **steady and then accelerated increase in 3-point attempts**, particularly from the mid-2010s onward. This shift reflects a league-wide strategic change, where teams increasingly prioritize spacing and perimeter shooting over traditional inside scoring.

Despite the rapid growth in shot volume, **three-point efficiency has remained relatively stable**, indicating that teams have not sacrificed shot quality for quantity. Instead, this suggests improved shot selection, better offensive systems, and the development of more skilled shooters across all positions.

At the team level, the analysis shows that **higher 3-point volume is often associated with stronger offensive performance**, especially when combined with above-average efficiency. This reinforces the idea that success in the modern NBA is closely linked to effective three-point shooting strategies, as an example **Stephen Curry** and the **Golden State Warriors** were once of the first team implementing this new trend and got great success in the late 2010s.

From a player perspective, the data highlights a clear evolution in roles. Positions that traditionally focused on interior play have increasingly incorporated perimeter shooting, demonstrating a shift toward a more **positionless and versatile style of basketball**.

Overall, the findings demonstrate that the three-point shot is no longer a complementary tactic but a **fundamental driver of offensive success** in today's NBA. As the game continues to evolve, the ability to effectively generate and convert three-point opportunities will remain a defining factor in competitive success across the NBA.

Limitations

While this analysis provides valuable insights into the evolution of three-point shooting in the NBA, several limitations should be considered.

First, the analysis focuses primarily on **aggregate team and player statistics**, which may overlook important contextual factors such as defensive pressure, game pace, or situational play (e.g., clutch moments). These elements can significantly influence both shot selection and efficiency.

Second, the study does not explicitly account for **rule changes and structural shifts in the league**, such as pace-of-play evolution or defensive regulations, which may have contributed to the increase in three-point attempts over time.

Finally, the **lack of spatial shot data** (e.g., shot location or shot quality metrics). Without this, it is difficult to fully assess whether improvements are driven by better shot selection or improved player skill.

Future Work

A natural next step would be to incorporate **advanced metrics**, such as effective field goal percentage (eFG%) or true shooting percentage (TS%), to better capture the efficiency impact of three-point shooting within overall scoring performance.

Including **shot location data** would allow for a more detailed exploration of shot selection, helping distinguish between contested and open shots, as well as identifying high-value scoring areas on the court.

Another valuable extension would be a **team-level case study approach**, analyzing specific teams that led or lagged behind the three-point revolution to better understand strategic differences and their outcomes.

The analysis could also be expanded to examine **player archetypes and positional evolution** in more depth, highlighting how different roles have adapted to the increasing importance of perimeter shooting.

In addition, incorporating **game context variables**—such as pace, opponent strength, or score differential—would provide a more nuanced understanding of when and why teams rely on three-point shooting.

Finally, developing **predictive models** could help forecast future trends in three-point usage and efficiency, offering actionable insights for teams looking to stay competitive as the game continues to evolve.

References

Data Sources:

- Kaggle. Source website for dataset acquisition.
- Basketball Reference. The original website from which the datasets were sourced and utilized to develop the glossary.

Tools:

- Excel (Data preparation)
- SQLite (Data Analysis)
- Tableau (Data visualisation and dashboard development)
- PowerPoint (Poster design and visualisation)
- GitHub (Project repository)

Additional Sources:

- History of the 3-Pointer. Publication that contributed to the conceptual framework for this analysis.

Images:

- NBA Logo obtained from clipartmax.com.
- Team's logos used on the dashboard obtained from stickPNG.com.

Use of AI

This project made limited use of AI tools, including ChatGPT, to support the writing and technical development process. AI assistance was used to refine explanations, improve clarity, and provide guidance when troubleshooting SQL queries.

In particular, AI was consulted to help identify potential errors and suggest alternative approaches when queries did not produce the expected results. All SQL code was reviewed, tested, and adapted by the author to ensure accuracy and alignment with the project objectives.

All data analysis, visualisations, and conclusions presented in this report are the original work of the author.

Acknowledgements

The author would like to acknowledge the feedback provided by Shawn James throughout the development of this project, which contributed to the refinement of the analysis, visualisations, and overall structure of the work.